

Large-Scale and Multi-Structured Databases

Project Design

BoardVerse

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Application Highlights

BoardVerse is a web application for board game players, providing a hub to discover games, engage with other players, share interests, and join exciting tournaments.

Main Features:

- **Extensive Board Game Catalog:** Explore detailed descriptions, reviews and ratings for a wide variety of games.
- **Game Reviews and Ratings:** Share feedback and guide others with your opinions.
- **Community Forum:** Connect, discuss, and engage with other board game enthusiasts.
- **Tournaments Management:** Create and join board games tournaments easily.

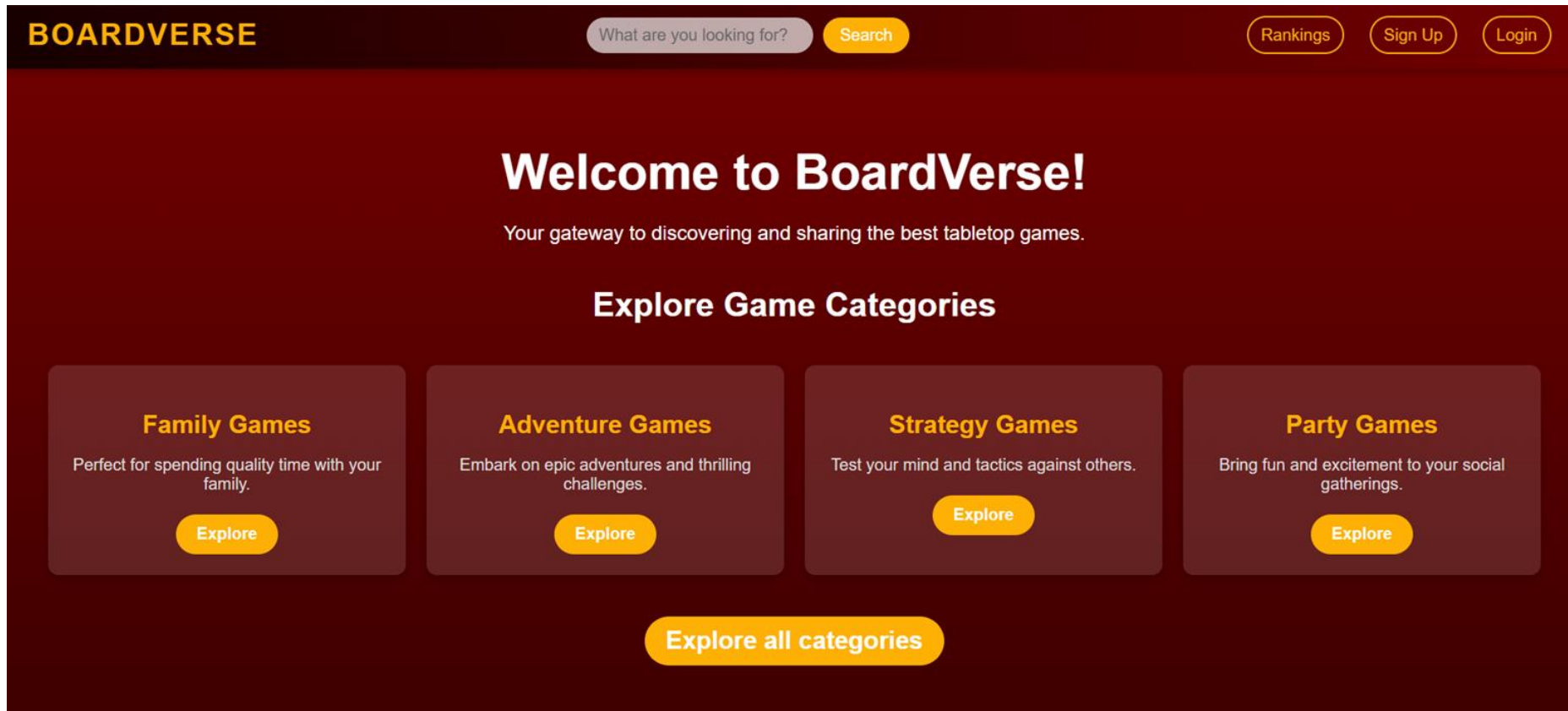
Actors and main mock-ups (i)

Actors:

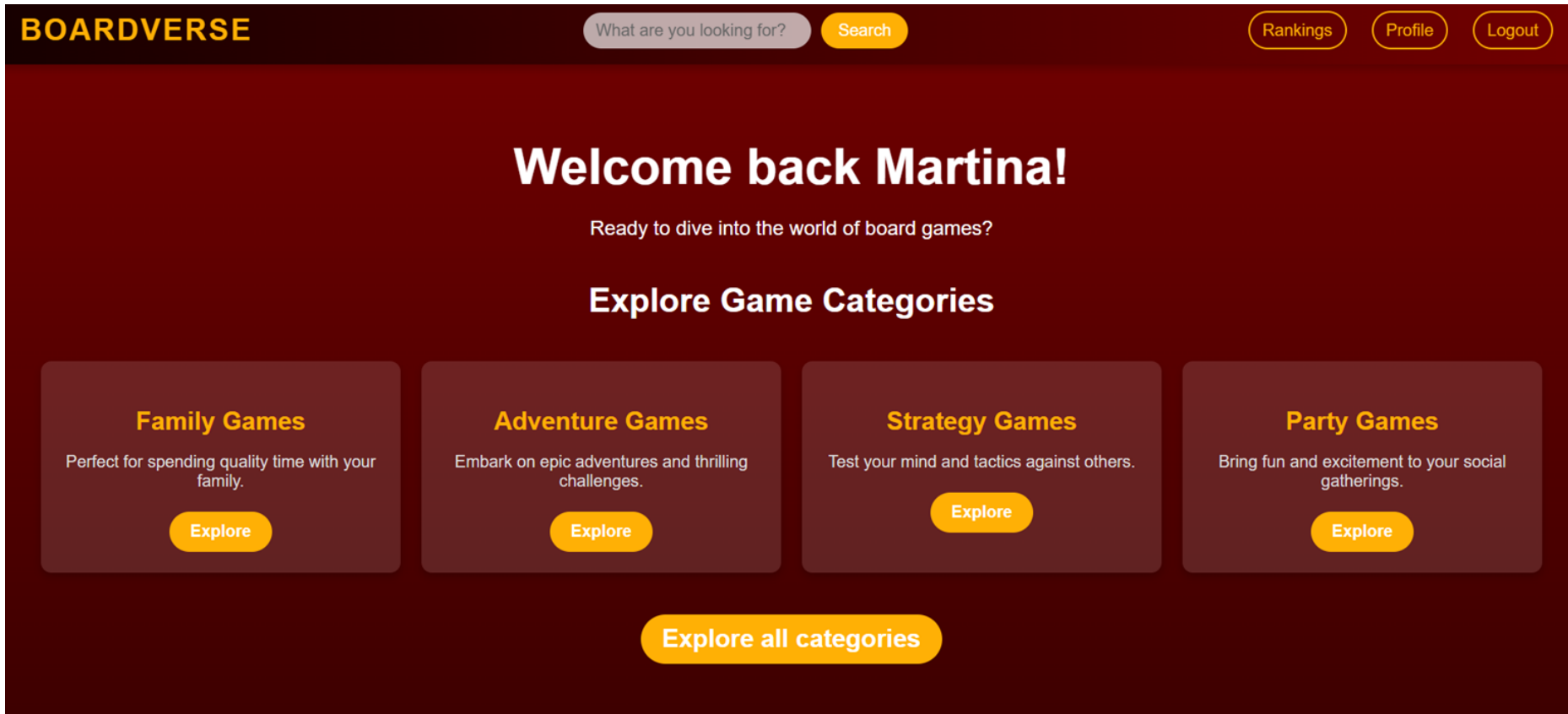
- *Unregistered User*

- *Administrator*

- *Registered User*



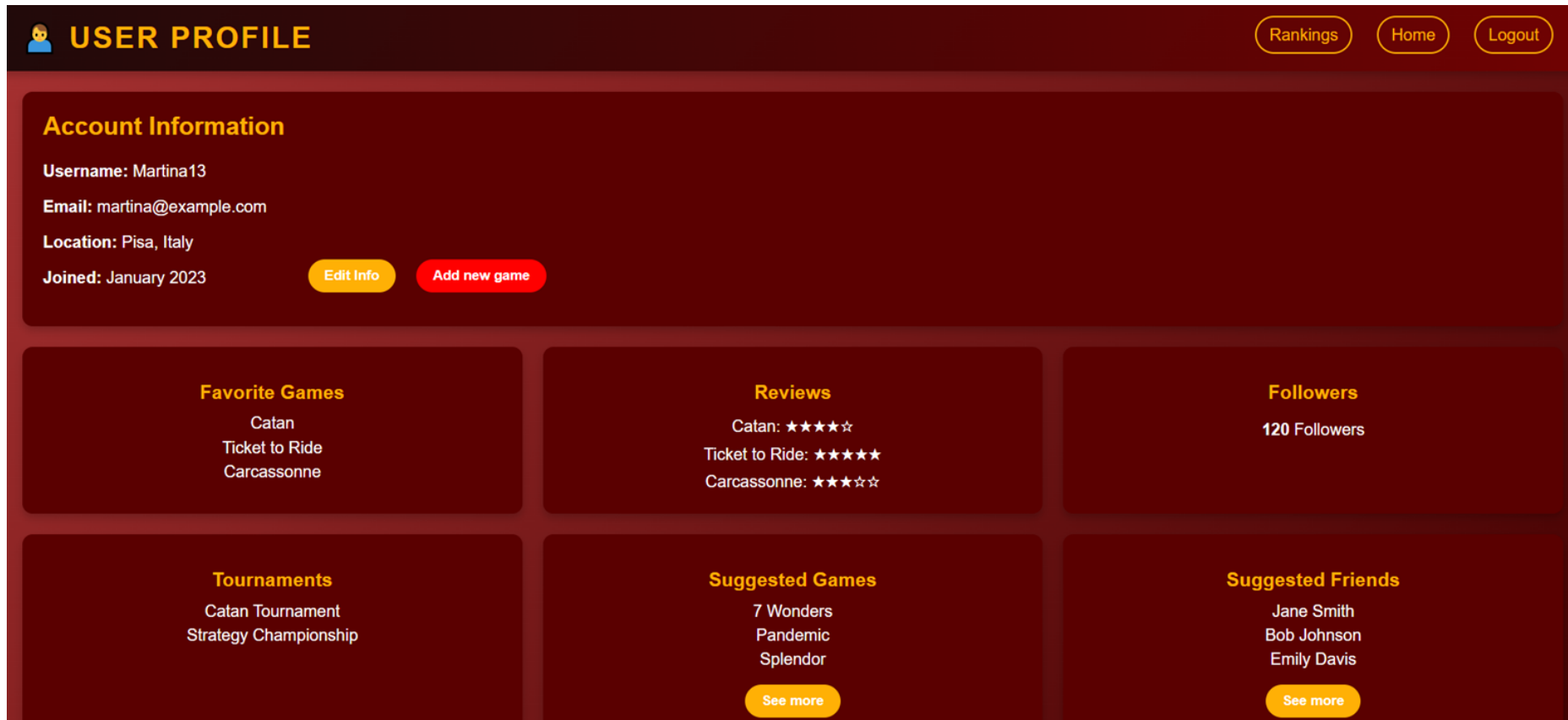
Actors and main mock-ups (ii)



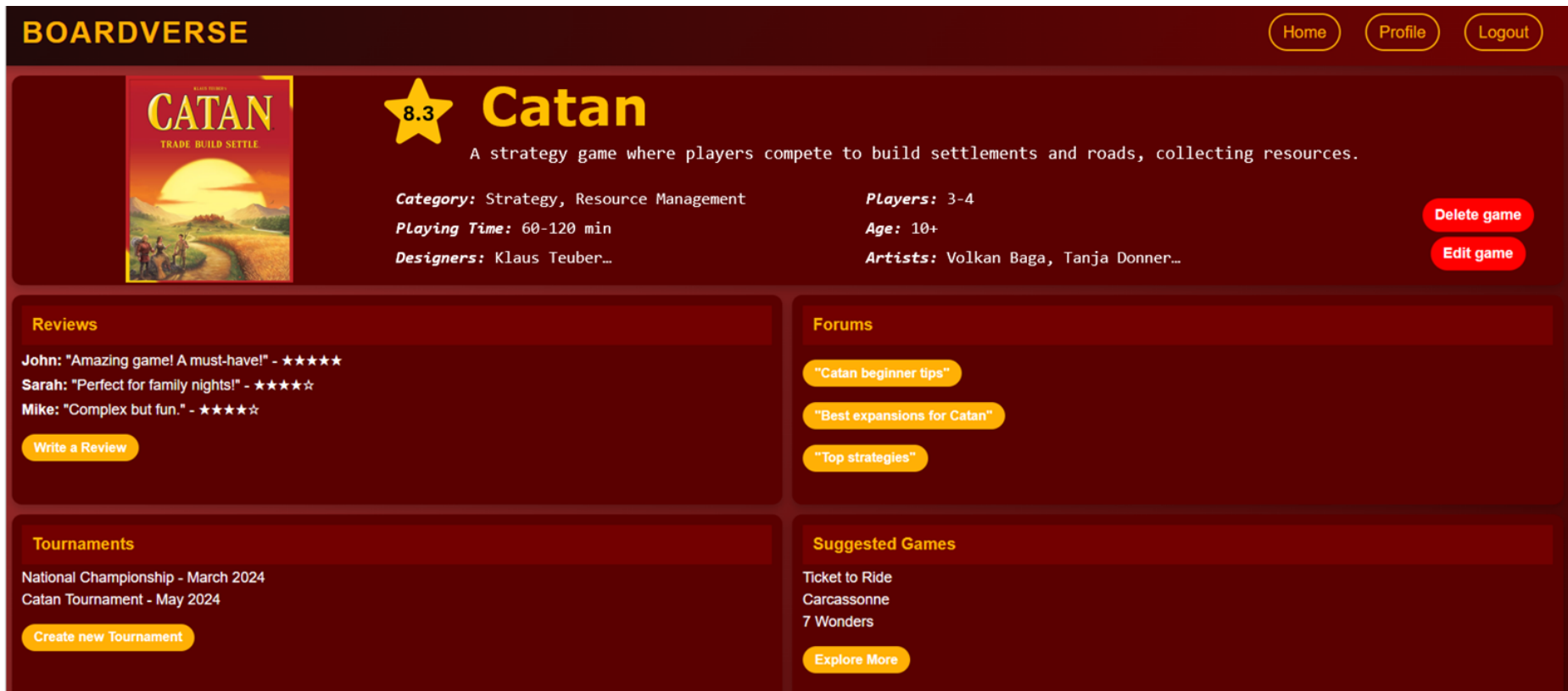
Actors and main mock-ups (iii)

TOP 10 BOARD GAMES			Home	Profile	Logout
Select Category:			Global		
1.	 Catan	Rating: 9.2			
2.	 Carcassonne	Rating: 9			
3.	 Ticket to Ride	Rating: 8.9			
4.	Pandemic	Rating: 8.8			
5.	7 Wonders	Rating: 8.7			
6.	Dominion	Rating: 8.6			
7.	Codenames	Rating: 8.5			
8.	Gloomhaven	Rating: 8.4			
9.	Azul	Rating: 8.3			
10.	Scythe	Rating: 8.2			

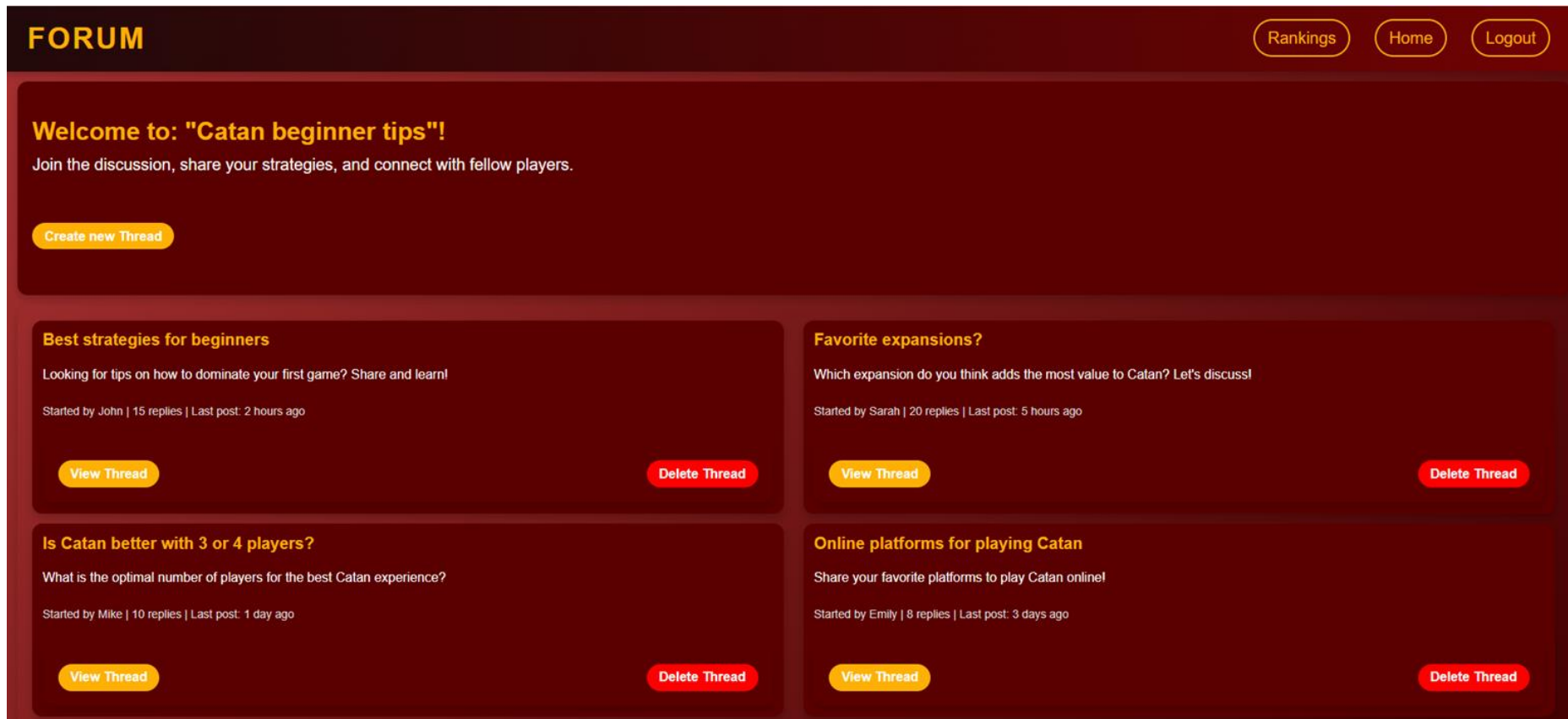
Actors and main mock-ups (iv)



Actors and main mock-ups (v)



Actors and main mock-ups (vi)



Actors and main mock-ups (vii)



Dataset Description

Source: Two platform, **BoardGameGeek (BGG)** and **BoardGameArena (BGA)**. BGG offers an XML API to retrieve data, while BGA needs scraping techniques, as no dedicated APIs are available.

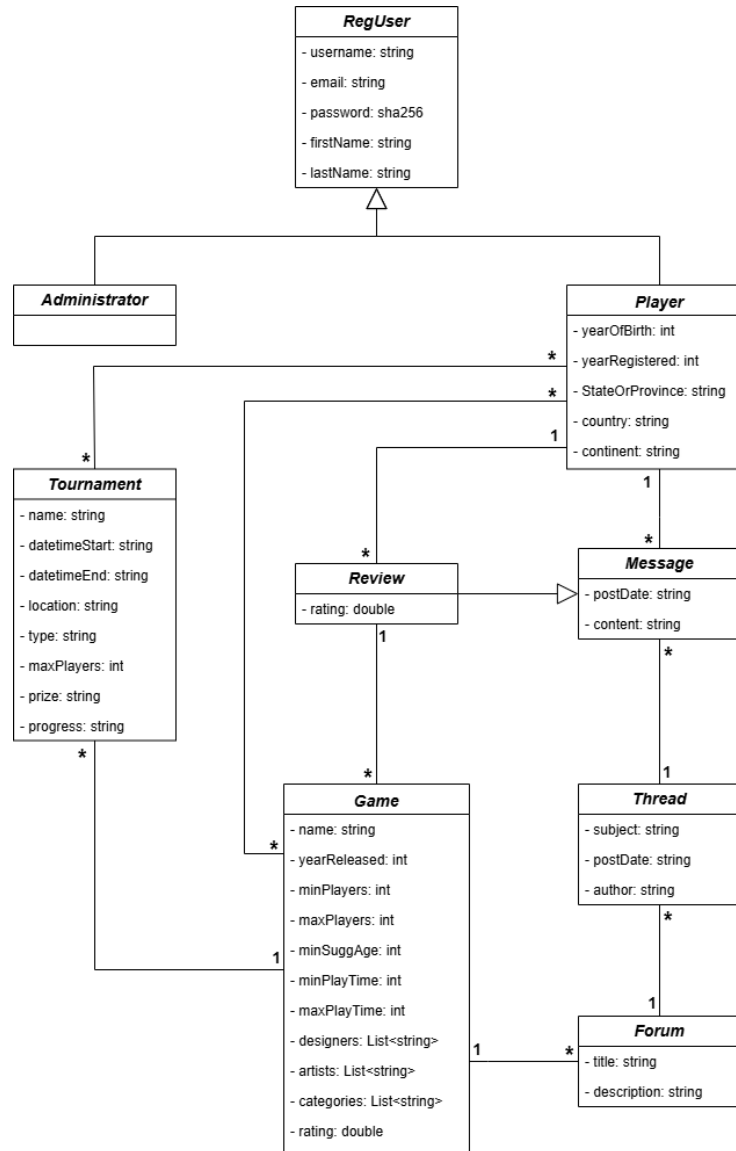
Description: The dataset will include information about games, reviews, forums, users (**BGG**) and tournaments (**BGA**).

Volume: Approximately **500 MB**, gathered through targeted scraping of the top-k most popular games (around **2,000 games** with associated metadata and related activities).

Variety: Achieved gathering on two different sites (**BGG, BGA**), with different communities and users.

Velocity/Variability: Data is nearly static. However, tournaments' data have dynamic importance for users, losing relevance once the tournament ends.

Preliminary UML Class Diagram



Relevant Queries and Entities handled by Document DB

Entities:

RegUser, Game, Review, Forum, Thread, Message, Tournament

The most relevant queries:

- Global board games ranking based on ratings, filtered by year and category
- Highest rated board games per location
- Most competitive games based on tournaments
- Hottest games based on forum activity
- Weighted rating based on user profiles

Requirements and Entities handled by Graph DB

Entities:

RegUser, Game, Tournament

The most relevant queries:

- *Users recommendation based on friends' friendships.*
- *Board games recommendation based on the user's favorite board games and friends' favorite board games.*
- *Tournament recommendation based on the tournaments that the user's friends are participating in.*

